CSC426 PRACTICAL ASSIGNMENT REPORT

COVER PAGE

Course Code: CSC426

Course Title: Web Application Development / Software Engineering Practical

Assignment Title: Development of a Simple Calculator and Login Authentication System

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1. INTRODUCTION

This practical assignment involved the development of two software applications using HTML, CSS, and JavaScript. The first application is a Simple Calculator capable of performing basic arithmetic operations such as addition, subtraction, multiplication, division, modulus, and exponentiation. The second application is a Login Authentication System that validates user credentials and provides appropriate feedback messages.

The objective of this assignment was to demonstrate an understanding of front-end web development concepts, user interface design, event handling, input validation, and application deployment using modern web technologies.

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2. TECHNOLOGIES USED

The following technologies were used during the development of the applications:

- HTML5 for structuring web pages.
- CSS3 for styling and responsive user interface design.
- JavaScript for implementing application logic and user interaction.
- GitHub for source code management and version control.
- Vercel for application deployment and hosting.

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3. SIMPLE CALCULATOR APPLICATION

3.1 Overview

The calculator application was developed to perform common mathematical operations through a graphical user interface. Users can enter numbers and operators using on-screen buttons and obtain calculated results instantly.

3.2 Features

The calculator supports the following operations:

- Addition (+)
- Subtraction (-)
- Multiplication (*)
- Division (/)
- Modulus (%)
- Exponent (^)
- Clear Function (C)

3.3 Functional Description

The calculator accepts user input through interactive buttons. The entered expression is displayed on the screen and evaluated when the equals (=) button is pressed. The Clear (C) button removes all current inputs and resets the calculator.

3.4 Screenshot

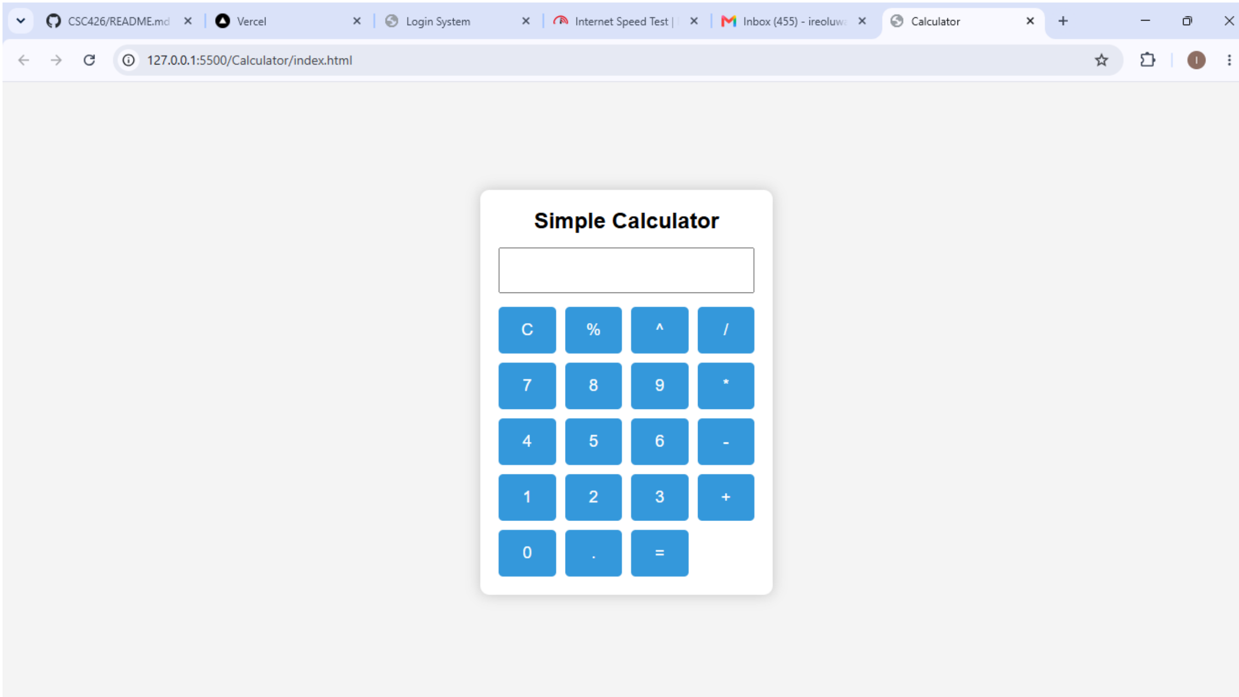
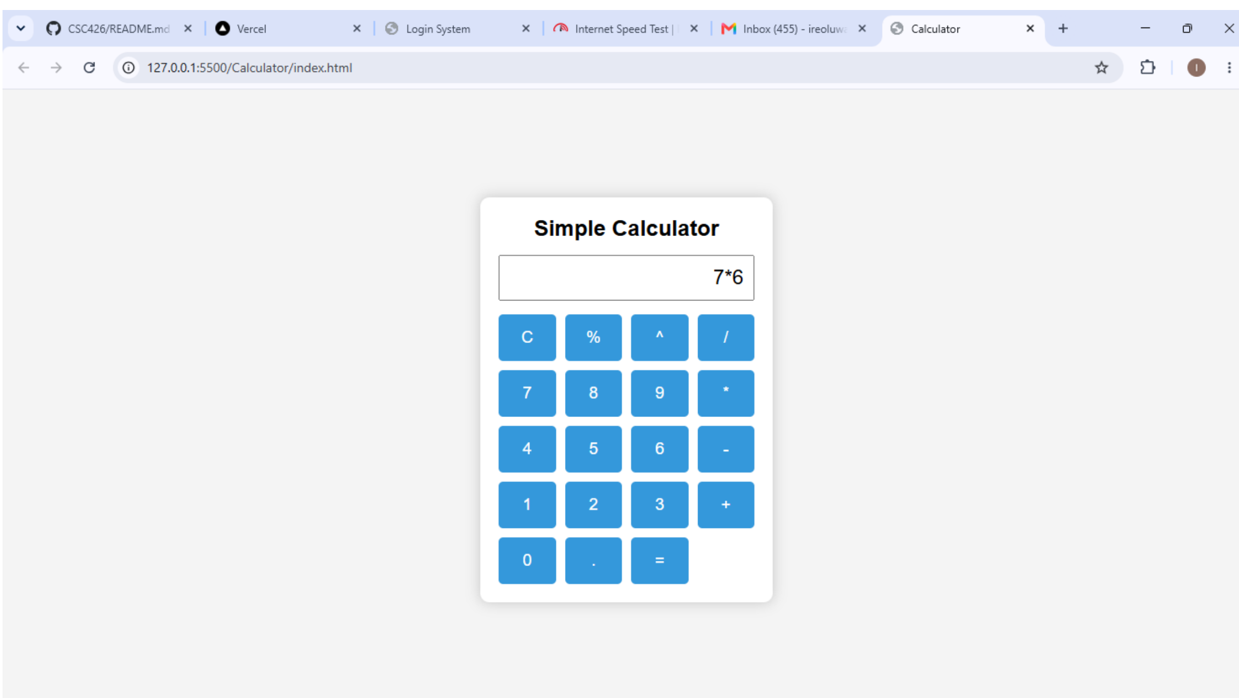


Figure 3.1 Calculator Interface



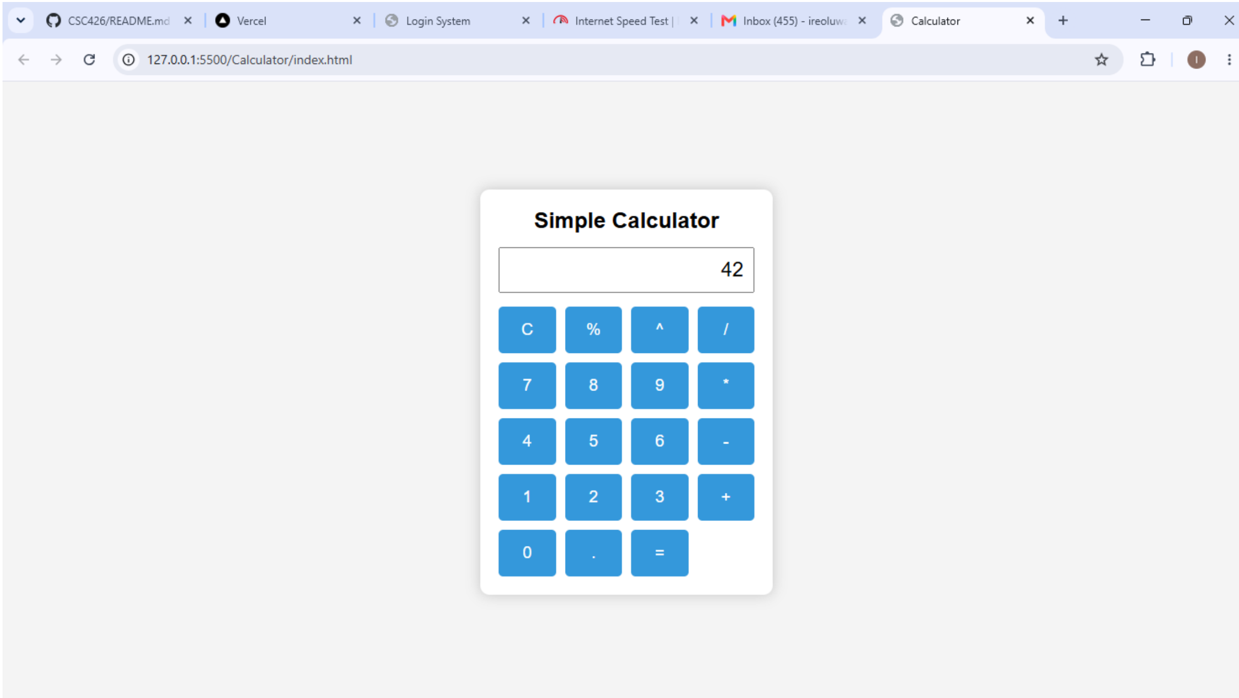


Figure 3.2 Calculator Outputs

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4. LOGIN AUTHENTICATION APPLICATION

4.1 Overview

The Login Authentication Application provides a secure and user-friendly interface for validating user credentials. The application checks user input and displays appropriate success or error messages based on the entered username and password.

4.2 Features

The login system contains the following features:

- Username Field
- Password Field

- Login Button
- Reset Button
- Input Validation
- Success Messages
- Error Messages
- Responsive User Interface

4.3 Functional Description

The application requires users to enter a username and password. Before authentication, the system validates that all fields have been completed. If any field is empty, an error message is displayed.

When valid credentials are entered, a success message is shown. If invalid credentials are entered, the system displays an appropriate error message informing the user of the failed login attempt.

4.4 Screenshot

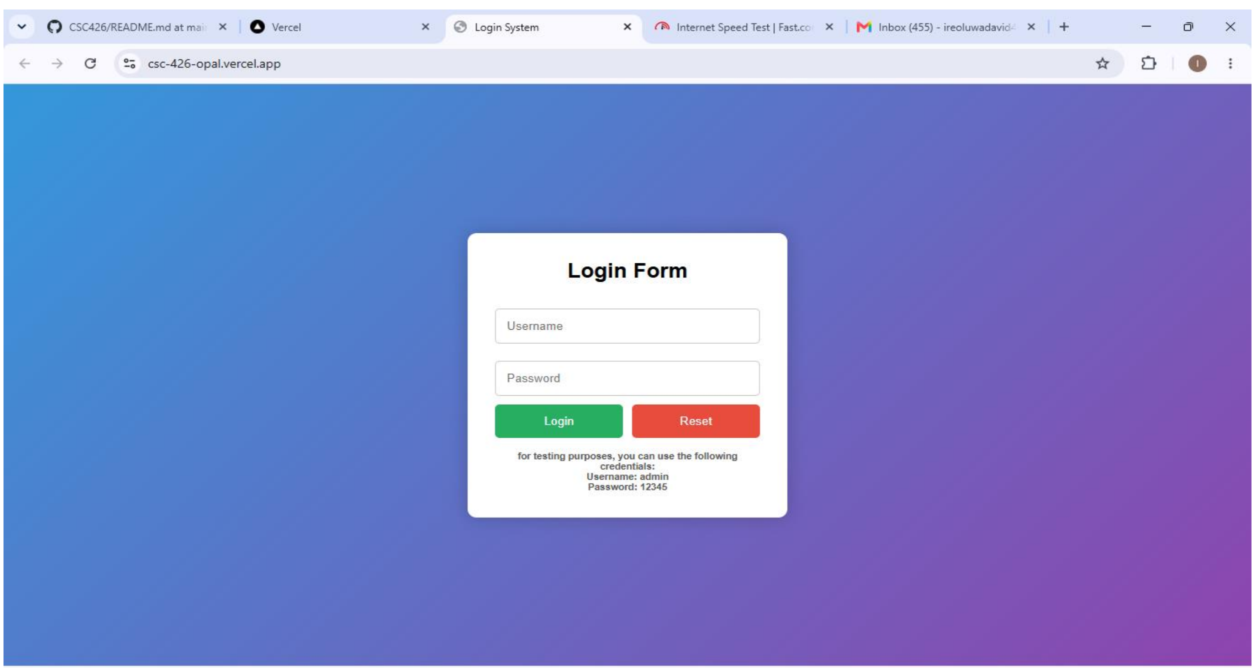

Figure 4.1 Login page

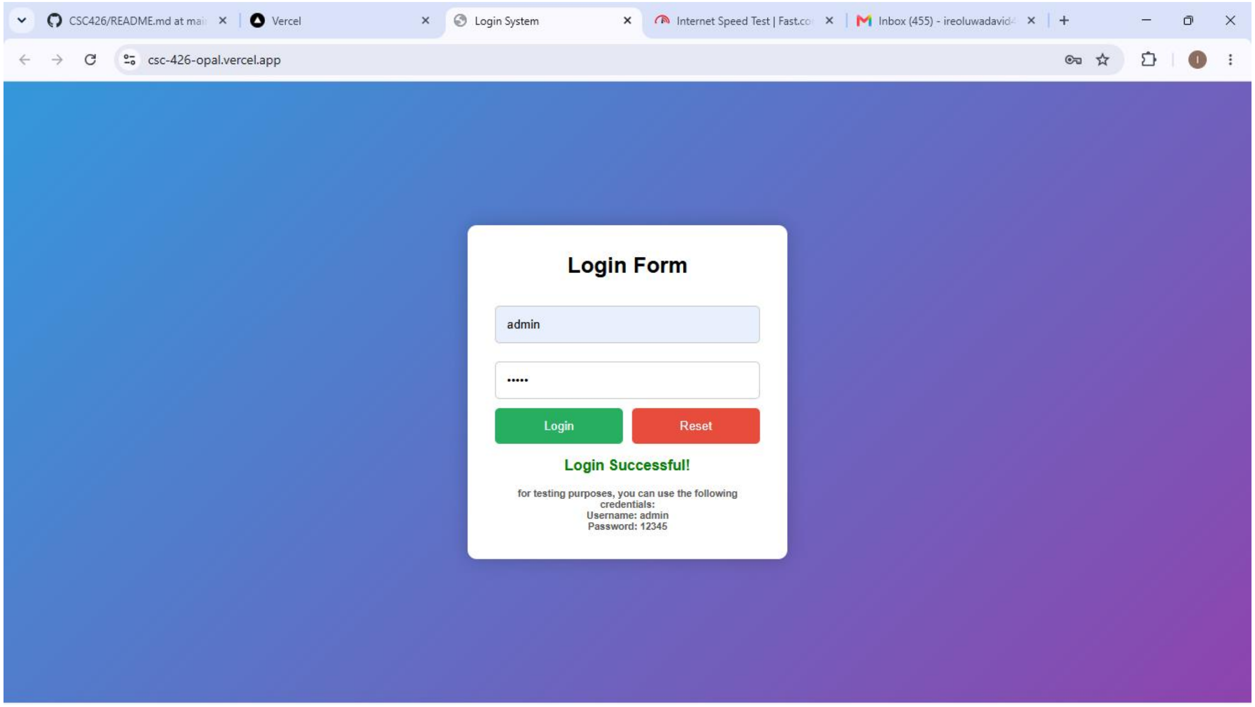

Figure 4.2 Successful Login

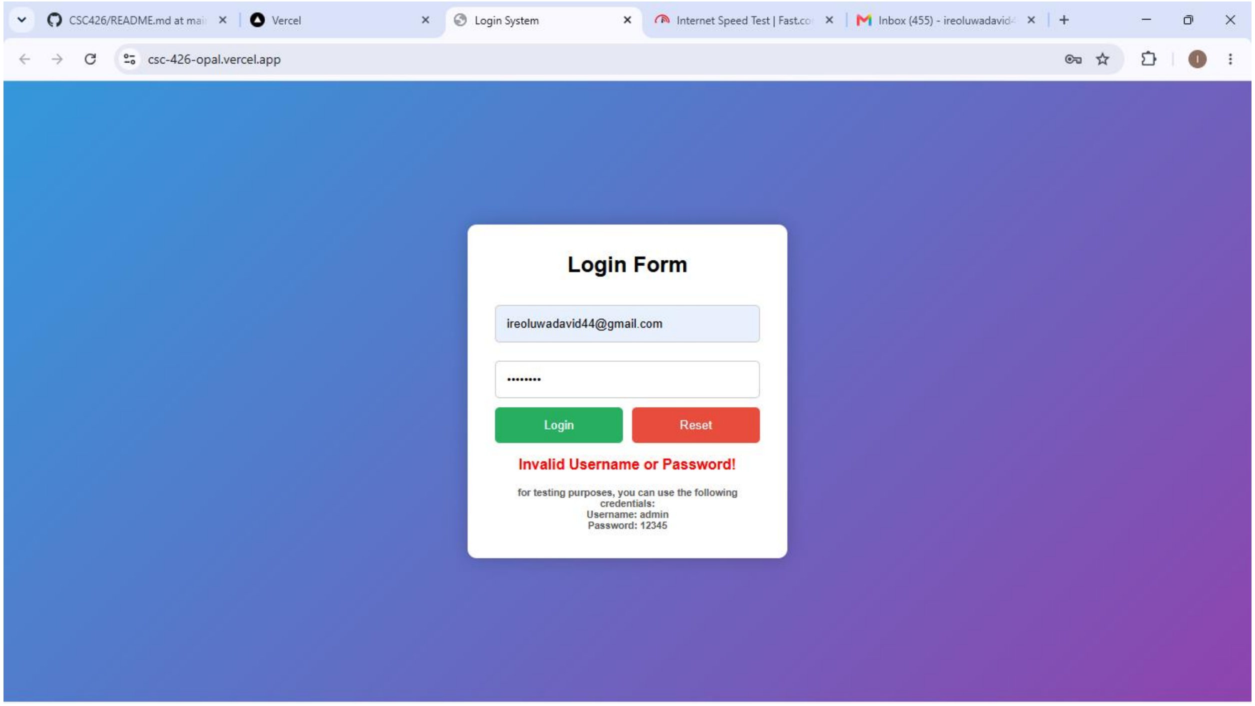

Figure 4.3 Successful Login

5. CHALLENGES ENCOUNTERED

During the development process, several challenges were encountered, including:

- Designing a responsive user interface that works across different screen sizes.
- Implementing input validation to prevent invalid data submission.
- Handling user interactions through JavaScript event listeners.
- Organizing project files within a single GitHub repository.

These challenges were successfully resolved through testing and debugging.

6. CONCLUSION

This practical assignment successfully demonstrated the development of two functional web applications using HTML, CSS, and JavaScript. The calculator application performs arithmetic computations efficiently, while the login authentication system validates user credentials and provides meaningful feedback.

The project enhanced understanding of front-end web development, user interface design, input validation, event-driven programming, version control using GitHub, and web application deployment using modern hosting platforms.

The source code, documentation, and application files have been uploaded to GitHub, while the Login Authentication Application has been deployed online for public access and testing.